

A Static Model of League Entry under Monopoly

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1 Benchmark: franchise supply and aggregate league value

I model the market for franchise memberships with a downward-sloping inverse demand curve. Imagine ranking prospective owner–market pairs by their willingness to pay for one membership, from highest to lowest. For n memberships, $P(n)$ represents the willingness to pay of the marginal accepted owner (the n th in this ranking). The benchmark treats n as continuous to obtain analytical conditions, although actual teams are indivisible. I specify this smooth approximation to inverse demand as

$$P(n) = Ae^{-Bn^\gamma}, \quad A > 0, \quad B > 0, \quad \gamma \geq 1, \quad n \geq 0. \quad (1)$$

The variables and parameters have the following meanings:

- n Number of franchise memberships supplied, treated as continuous for the derivations.
- A Scale of willingness to pay; mathematically, the demand intercept $P(0) = A$.
- B Rate at which the marginal prospective owner’s willingness to pay declines across the ranking of markets.
- γ Curvature parameter. At $\gamma = 1$, moving to the next market reduces marginal willingness to pay by the same proportion. At $\gamma > 1$, that proportional reduction rises with league size.

Both A and $P(n)$ are measured in money per membership; B has units of teams raised to $-\gamma$ ¹. The marginal valuation remains strictly positive at every finite n and approaches zero as n grows. As more slots become available, the market-clearing price falls and the league reaches less valuable prospective markets. In this benchmark, each owner’s underlying willingness to pay is fixed across alternative league sizes. The market price of existing memberships can nevertheless fall as additional rights become available; separate changes in incumbents’ operating cash flows are not modeled.

To value the league’s membership rights, assume that they are comparable claims priced at the common market-clearing price $P(n)$ when the league has n teams. The marginal prospective owner’s willingness to pay determines that price. Marking all n memberships to the resulting price gives their aggregate market value:

$$V(n) = nP(n) = Ane^{-Bn^\gamma}. \quad (2)$$

The expression $V(n)$ is a valuation of the stock of membership rights, not expansion-fee receipts or the league’s annual operating revenue. It includes the change in the market price assigned to existing rights as supply grows. It assumes a common market price for comparable memberships; with city-specific franchise values, the corresponding aggregate valuation would instead sum those individual values.

¹The exponent Bn^γ must be dimensionless. Since n is measured in teams, n^γ has units of teams ^{γ} . Therefore, B has units of teams ^{$-\gamma$} ; for example, when $\gamma = 2$, its units are teams ^{-2} .

2 Cost of supplying franchise memberships

As the league grows, an additional team requires baseline integration and administration and may increase scheduling, travel coordination, and governance complexity. A parsimonious cost specification with increasing marginal cost is

$$C(n) = c_0 n + \frac{c_1}{2} n^2, \quad c_0 > 0, \quad c_1 > 0, \quad (3)$$

with marginal cost

$$MC(n) = C'(n) = c_0 + c_1 n. \quad (4)$$

Here c_0 is the baseline real resource cost per membership, measured in money per team; c_1 measures how much the marginal resource cost rises with league size, in money per team squared. An expansion fee is a payment by a new owner, not a resource cost. Likewise, costs paid directly by owners and already incorporated into the valuation $P(n)$ should not be counted again in $C(n)$. Fixed costs of operating an existing league are sunk for this entry decision. I assume $A > c_0$ to obtain positive, finite interior choices.

3 Monopoly entry

The league chooses n to maximize aggregate membership market value net of the real cost of supplying slots. This benchmark abstracts from how that value is divided between original owners and new owners:

$$\max_{n \geq 0} \Lambda(n) = A n e^{-B n^\gamma} - c_0 n - \frac{c_1}{2} n^2. \quad (5)$$

The marginal contribution to aggregate membership value and the derivative of net league value are

$$MV(n) = V'(n) = A e^{-B n^\gamma} (1 - B \gamma n^\gamma), \quad (6)$$

$$\Lambda'(n) = A e^{-B n^\gamma} (1 - B \gamma n^\gamma) - (c_0 + c_1 n). \quad (7)$$

The monopoly choice n_M therefore satisfies

$$A e^{-B n_M^\gamma} (1 - B \gamma n_M^\gamma) = c_0 + c_1 n_M. \quad (8)$$

Equation (8) defines n_M implicitly and can be solved numerically. The solution is unique on $0 < n < (B\gamma)^{-1/\gamma}$: marginal aggregate value falls from A to zero on this interval, while marginal cost rises from c_0 ; $A > c_0$ guarantees one crossing. The upper endpoint maximizes aggregate membership market value before supply costs, so positive marginal cost implies $n_M < (B\gamma)^{-1/\gamma}$.

4 Efficient entry in the franchise-membership market

For a partial-equilibrium efficiency benchmark, assume $P(q)$ measures the fixed marginal gross value of the q th membership in the ranking and that entry creates no other benefits or costs. Expansion fees are transfers from new owners to the league, so they do not themselves create total surplus. The combined surplus of owners and the league is

$$W(n) = \int_0^n P(q) dq - C(n) = A \int_0^n e^{-B q^\gamma} dq - c_0 n - \frac{c_1}{2} n^2. \quad (9)$$

Efficiency requires the prospective owner's willingness to pay for an additional slot to equal the real marginal cost of supplying that slot:

$$W'(n_E) = P(n_E) - MC(n_E) = 0. \quad (10)$$

The efficient entry level n_E is the solution to

$$Ae^{-Bn_E^\gamma} = c_0 + c_1 n_E. \quad (11)$$

The left-hand side falls with n and the right-hand side rises, so $A > c_0$ guarantees a unique positive solution. For general γ , compute n_E numerically. When $\gamma = 1$, it can be expressed with the principal Lambert W function²:

$$n_E = \frac{1}{B} W_0 \left(\frac{AB}{c_1} e^{Bc_0/c_1} \right) - \frac{c_0}{c_1}, \quad (\gamma = 1). \quad (12)$$

This is efficiency in the market for franchise membership. A fall in the common market price alone transfers value between prospective buyers and sellers; if entry also changes incumbents' underlying cash flows or willingness to pay, the welfare calculation in equation (9) must include those real effects separately. Fan surplus and player effects are also outside this benchmark.

5 Comparison of monopoly and efficient entry

Because all n memberships are valued at the common market price $P(n)$, their aggregate market value is $V(n) = nP(n)$. A marginal increase in league size has two effects: it adds a membership valued at $P(n)$, and it lowers the price used to value the existing n memberships. Differentiating $V(n)$ captures both effects:

$$MV(n) \equiv V'(n) = P(n) + nP'(n) < P(n), \quad n > 0, \quad (13)$$

since $P'(n) < 0$. The term $nP'(n)$ is the change in the existing memberships' aggregate market valuation; it is not an additional cost of supplying a franchise.

The league chooses n_M where this marginal change in aggregate market value equals marginal supply cost: $MV(n_M) = MC(n_M)$. The membership-market efficiency benchmark instead compares the next prospective owner's willingness to pay with that cost. Thus $W'(n) = P(n) - MC(n)$, and at the league's chosen size,

$$W'(n_M) = P(n_M) - MC(n_M) = -n_M P'(n_M) = AB\gamma n_M^\gamma e^{-Bn_M^\gamma} > 0. \quad (14)$$

$-n_M P'(n_M) > 0$ shows that a marginal increase in league size would still raise membership-market surplus at n_M .

Moreover, $P(n) - MC(n)$ is strictly decreasing because $\frac{d}{dn}[P(n) - MC(n)] = P'(n) - c_1 < 0$. It starts at $A - c_0 > 0$ and reaches zero exactly once, at n_E . Since it remains positive at n_M , we obtain

$$\boxed{n_M < n_E \quad \text{for } A > c_0 > 0, B > 0, \gamma \geq 1, c_1 > 0.} \quad (15)$$

Hence, in the continuous benchmark, a league maximizing aggregate membership market value net of supply cost chooses fewer slots than the membership-market efficiency benchmark. This comparison does not include effects on fans, players, or incumbent teams' underlying cash flows.

Figure 1 plots the two marginal conditions. The league chooses the intersection of $MV(n)$ and $MC(n)$; efficient entry occurs farther to the right, at the intersection of $P(n)$ and $MC(n)$. The numerical values in the figure illustrate the model and are not calibrated estimates.

²The Lambert W function is defined implicitly by $W(z)e^{W(z)} = z$. The principal branch, denoted W_0 , is the branch satisfying $W_0(z) \geq -1$ for real $z \geq -e^{-1}$; in particular, it provides the unique real solution for $z > 0$.

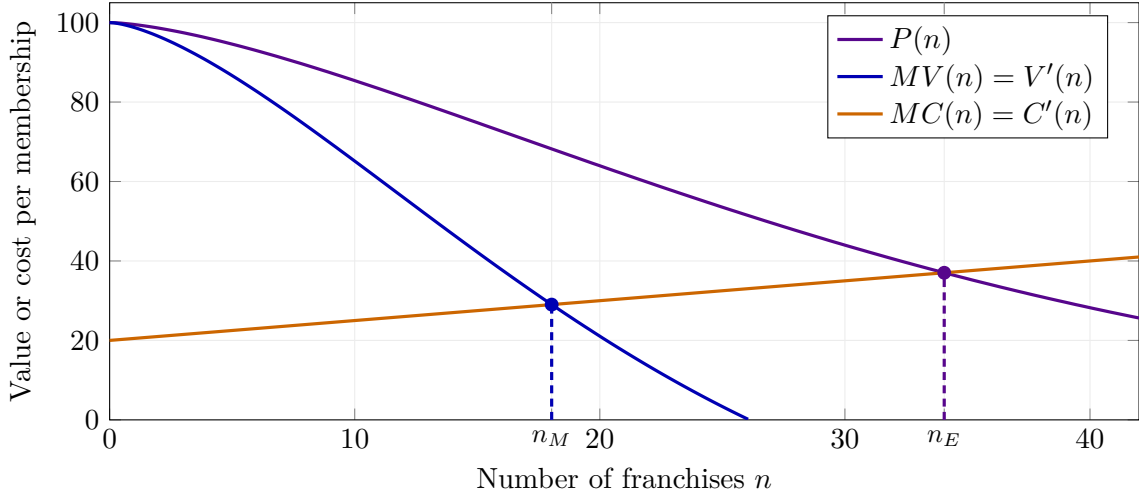


Figure 1: Illustrative private and efficient entry levels for $A = 100$, $B = 0.005$, $\gamma = 1.5$, $c_0 = 20$, and $c_1 = 0.5$ (normalized monetary units). The intersections imply $n_M \approx 18.03$ and $n_E \approx 34.05$ in the continuous approximation.

Figure 2 highlights the welfare consequence of this restriction in entry. Because the league stops at $n_M < n_E$, there is a range of additional franchise slots for which prospective owners' willingness to pay, $P(n)$, still exceeds the league's marginal cost of supplying those slots, $MC(n)$. The resulting deadweight loss is the area between $P(n)$ and $MC(n)$ over the interval $[n_M, n_E]$:

$$DWL = \int_{n_M}^{n_E} [P(n) - MC(n)] dn.$$

This area represents the unrealized gains from mutually beneficial entry that are forgone when the league restricts membership below the efficient level.

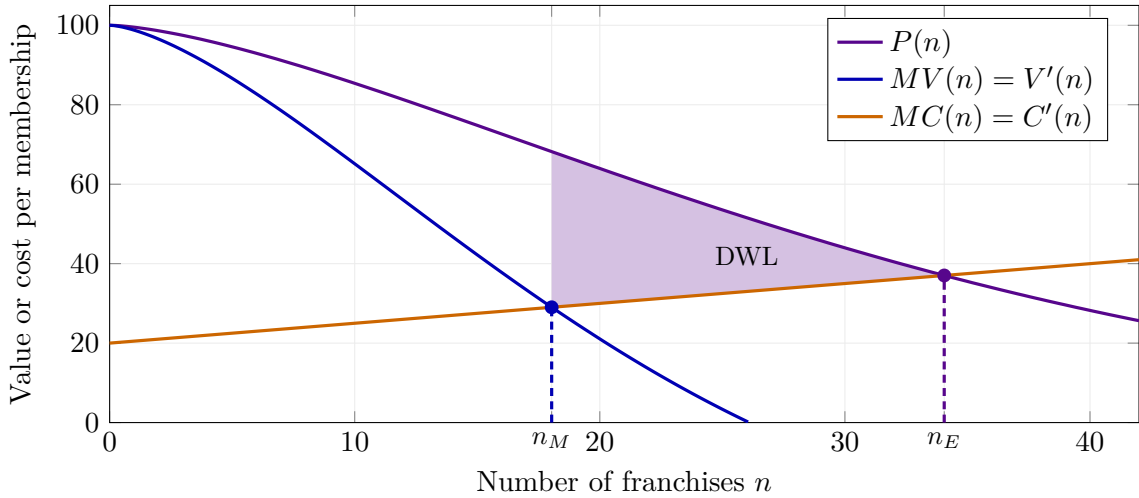


Figure 2: Deadweight loss within the franchise-membership benchmark. The shaded area equals $DWL = W(n_E) - W(n_M) = \int_{n_M}^{n_E} [P(q) - MC(q)] dq \approx 301.46$ for the illustrative parameters $A = 100$, $B = 0.005$, $\gamma = 1.5$, $c_0 = 20$, and $c_1 = 0.5$.

Actual franchise counts are integers. For a discrete application, compare $\Lambda(n+1) - \Lambda(n)$ with zero for the league's private decision and $W(n+1) - W(n)$ with zero for the efficient decision. The continuous strict inequality can become a weak inequality after rounding.